Prerequisite: Blockchain Development Environment

Practical No. 0

Installation and Configuration of Blockchain Development Environment

Status: Prerequisite setup — evaluated as a completion check (not mapped to a CO)

Aim

To install and configure the essential software tools (Node.js, Ganache, MetaMask, Remix IDE) required for blockchain and smart-contract development, and to verify that the setup works correctly.

Theory

A blockchain development environment typically consists of a local test blockchain (Ganache) that simulates an Ethereum network on the developer's machine, a browser wallet (MetaMask) used to manage accounts and sign transactions, a smart-contract IDE (Remix) used to write, compile and deploy Solidity contracts, and a JavaScript runtime (Node.js) used to run supporting scripts and package managers such as npm. Setting up this toolchain correctly is the foundation for every subsequent practical in this course, since almost all Ethereum-based experiments (Units III–V) depend on these tools being installed and connected to each other.

Note: *MetaMask's interface updates frequently; menu labels and screen layouts may differ slightly from any screenshots, but the underlying steps remain the same.*

Requirements & Software

Field	Details
Hardware / System Requirement	• PC/Laptop with Windows, Linux, or macOS; minimum 4 GB RAM • Active internet connection • Google Chrome or Mozilla Firefox browser
Software Required	• Node.js (v18 or later) — https://nodejs.org • Ganache (v2.x) — https://trufflesuite.com/ganache • MetaMask browser extension — https://metamask.io • Remix IDE — used online at https://remix.ethereum.org (no installation required)

Procedure / Steps to Run

1. Download and install Node.js from the official website. Verify the installation by opening a terminal / command prompt and running `node -v` and `npm -v` .
2. Download and install Ganache. Launch it and click 'Quickstart' to start a local Ethereum test blockchain. Note the RPC Server address (usually http://127.0.0.1:7545) and the list of 10 pre-funded accounts, each with 100 test ETH. Keep the Ganache window open throughout the lab session — MetaMask and Remix can only communicate with Ganache while it is actively running.
3. Install the MetaMask extension in your browser and create a new wallet (store the seed phrase safely — for testing only).

4. In MetaMask, add a new network manually with RPC URL http://127.0.0.1:7545, Chain ID 1337, and currency symbol ETH. When the RPC URL is entered correctly, the Chain ID field fills automatically — this confirms MetaMask has connected to Ganache.
5. Copy the private key of one Ganache account (click the key icon next to an account). In MetaMask, open the account menu → 'Add wallet' → 'Private Key' (or 'Import Account'), paste the key, and import it. Confirm that the 100 ETH balance appears while MetaMask is on the Ganache network.
6. Open https://remix.ethereum.org and confirm the IDE loads with the default workspace and sample contracts.
7. (For later practicals) In MetaMask, enable 'Show test networks' under Settings → Advanced, and confirm the Sepolia test network appears in the network dropdown. Test ETH for Sepolia can be obtained from a faucet later when required.

Troubleshooting — Common Issues

"Could not fetch chain ID / RPC URL" error when saving the network: Ensure the RPC URL includes the prefix http:// (i.e., http://127.0.0.1:7545, not 127.0.0.1:7545) and that Ganache is running before saving. Once the URL is correct, the Chain ID auto-fills, confirming the connection.

Chain ID mismatch: Some Ganache versions use Chain ID 5777 instead of 1337. Use whatever value Ganache reports under Settings → Server; when the RPC URL is correct, MetaMask fetches it automatically.

Imported account shows 0 ETH: This almost always means MetaMask is on the wrong network (e.g., Ethereum Mainnet). Switch the network dropdown to 'Ganache Local'. Also confirm Ganache is still running — closing and restarting it with Quickstart generates a fresh chain with different accounts.

Cannot find 'Import Account': In newer MetaMask versions the import option is under the account menu → 'Add wallet' → 'Private Key'. The '+ Add account' button only creates new empty accounts and will not import the Ganache key.

Expected Output / Observation

Node.js and npm versions are displayed in the terminal. Ganache shows a running local blockchain with 10 accounts, each holding 100 ETH. MetaMask displays the imported account with the same 100 ETH balance while on the Ganache network, confirming that MetaMask is correctly connected to the local Ganache blockchain. Remix IDE opens successfully in the browser.

Outcome: *A fully configured environment in which Ganache, MetaMask, and Remix are interconnected and ready for smart-contract development in the practicals that follow.*